

More Angular Momentum

$$[\hat{J}_i, \hat{J}_j] = i\hbar \epsilon_{ijk} \hat{J}_k \rightarrow \text{Angular Momentum is a triplet } \hat{J}_i \text{ of Hermitian linear operators on some complex vector space } V.$$

$$\hat{J}_i \equiv \hat{J}_i^{(1)} \otimes 1 + 1 \otimes \hat{J}_i^{(2)}$$

Sum of angular momentum is an angular momentum in the tensor product.

$\Rightarrow$ Say for non-zero constant α & β

$$\tilde{J}_i \equiv \alpha \hat{J}_i^{(1)} \otimes 1 + \beta 1 \otimes \hat{J}_i^{(2)}$$

$\hookrightarrow$ This is NOT an angular momentum

$$[\tilde{J}_i, \tilde{J}_j] = i\hbar \epsilon_{ijk} (\alpha^2 \hat{J}_k^{(1)} \otimes 1 + \beta^2 1 \otimes \hat{J}_k^{(2)})$$

As this does not satisfy our commutation relation

The only solution is $\alpha^2 = \alpha \rightarrow \alpha = 1$
 $\beta^2 = \beta \rightarrow \beta = 1$

$\Rightarrow$ Adding two $\hat{S} (1/2)$:

$$S^2 |s, m\rangle = \hbar^2 s(s+1) |s, m\rangle$$

$$S_z |s, m\rangle = \hbar m |s, m\rangle$$

where,

$$s = \frac{1}{2} ; m = \pm \frac{1}{2}$$

$$\text{Basis : } |\frac{1}{2}, \frac{1}{2}\rangle \text{ \& } |\frac{1}{2}, -\frac{1}{2}\rangle$$

$$\cdot S^2 |+\rangle ; S^2 |-\rangle = \frac{3}{4} \hbar^2$$

$$\cdot \hat{S} |+\rangle = \frac{\hbar}{2} , \hat{S} |-\rangle = -\frac{\hbar}{2}$$

⇒ Consider two spin-Half particles :

$$\begin{aligned} \text{Total spin: } \hat{S}_i &\equiv \hat{S}_i^{(1)} \otimes 1 + 1 \otimes \hat{S}_i^{(2)} \\ &\equiv \hat{S}_i^{(1)} + \hat{S}_i^{(2)} \quad (\text{brevity}) \end{aligned}$$

$$En: \hat{S}_z = \hat{S}_z^{(1)} + \hat{S}_z^{(2)}$$

Recall: for $V_1 \otimes V_2$

$$\begin{aligned} \hookrightarrow \text{Basis: } & \left| \frac{1}{2}, \frac{1}{2} \right\rangle_1 \otimes \left| \frac{1}{2}, \frac{1}{2} \right\rangle_2 \\ & \left| \frac{1}{2}, \frac{1}{2} \right\rangle_1 \otimes \left| \frac{1}{2}, -\frac{1}{2} \right\rangle_2 \\ & \left| \frac{1}{2}, -\frac{1}{2} \right\rangle_1 \otimes \left| \frac{1}{2}, \frac{1}{2} \right\rangle_2 \\ & \nearrow \left| \frac{1}{2}, -\frac{1}{2} \right\rangle_1 \otimes \left| \frac{1}{2}, -\frac{1}{2} \right\rangle_2 \end{aligned}$$

Possibilities for these

- × 1. Four singlet: ($s=0$)
- × 2. Two doublet: ($s=1/2$)
- × 3. One doublet ($s=1/2$) & Two singlet ($s=0$)
- 4. One triplet ($s=1$) & one singlet ($s=0$)
- × 5. One $s=\frac{3}{2}$ Multiplet

Aligned: Triplet $s=1$ is 3-dim representation : $m = +1, 0, -1$

Antialigned: Singlet $s=0$ is 1-dim representation : $m = 0$

Doublet $s=\frac{1}{2}$: $m = \frac{1}{2}, -\frac{1}{2}$

↪ A doublet can occur in two particle system

Four singlets aren't possible as some have aligned spins.

So for 1-triplet (3 states: $m=\pm 1, 0$) & 1-singlet (1 state: $m=0$) we have total 4 states.

written as: $(s=\frac{1}{2}) \otimes (s=\frac{1}{2}) = (s=1) \oplus (s=0)$
or

$$\boxed{\frac{1}{2} \otimes \frac{1}{2} = 1 \oplus 0}$$

$$m=1 : |\frac{1}{2}, \frac{1}{2}\rangle_1 \otimes |\frac{1}{2}, \frac{1}{2}\rangle_2$$

$$m=0 : |\frac{1}{2}, \frac{1}{2}\rangle_1 \otimes |\frac{1}{2}, -\frac{1}{2}\rangle_2 ; |\frac{1}{2}, -\frac{1}{2}\rangle_1 \otimes |\frac{1}{2}, \frac{1}{2}\rangle_2$$

$$m=-1 : |\frac{1}{2}, -\frac{1}{2}\rangle_1 \otimes |\frac{1}{2}, -\frac{1}{2}\rangle_2$$

For $(s=1)$ triplet:

$$|s, m\rangle \begin{cases} |1, 1\rangle = |\frac{1}{2}, \frac{1}{2}\rangle_1 \otimes |\frac{1}{2}, \frac{1}{2}\rangle_2 \\ |1, 0\rangle = \alpha |\frac{1}{2}, \frac{1}{2}\rangle_1 \otimes |\frac{1}{2}, -\frac{1}{2}\rangle_2 + \beta |\frac{1}{2}, -\frac{1}{2}\rangle_1 \otimes |\frac{1}{2}, \frac{1}{2}\rangle_2 \\ |1, -1\rangle = |\frac{1}{2}, -\frac{1}{2}\rangle_1 \otimes |\frac{1}{2}, -\frac{1}{2}\rangle_2 \end{cases}$$

For $(s=0)$ singlet:

$$|0, 0\rangle = \gamma |\frac{1}{2}, \frac{1}{2}\rangle_1 \otimes |\frac{1}{2}, -\frac{1}{2}\rangle_2 + \delta |\frac{1}{2}, -\frac{1}{2}\rangle_1 \otimes |\frac{1}{2}, \frac{1}{2}\rangle_2$$

=> Now to find $\alpha, \beta, \gamma, \delta$:

recall:

$$J_{\pm}(j, m) = \hbar \sqrt{j(j+1) - m(m \pm 1)} |j, m \pm 1\rangle$$

$$J_- |1, 1\rangle = \hbar \sqrt{2} |1, 0\rangle$$

$$J_- |\frac{1}{2}, \frac{1}{2}\rangle = \hbar |\frac{1}{2}, -\frac{1}{2}\rangle$$

$$J_+ |\frac{1}{2}, -\frac{1}{2}\rangle = \hbar |\frac{1}{2}, \frac{1}{2}\rangle \rightarrow \text{since } J \text{ can be } S$$

$$S_- |1, 1\rangle = \left(S_-^{(1)} |\frac{1}{2}, \frac{1}{2}\rangle_1 \right) \otimes |\frac{1}{2}, \frac{1}{2}\rangle_2 + |\frac{1}{2}, \frac{1}{2}\rangle_1 \otimes \left(S_-^{(2)} |\frac{1}{2}, \frac{1}{2}\rangle_2 \right)$$

since

$$S = S^{(1)} + S^{(2)}$$

step
down

$$|1,0\rangle = \frac{1}{\sqrt{2}} \left(\left| \frac{1}{2}, \frac{1}{2} \right\rangle_1 \otimes \left| \frac{1}{2}, -\frac{1}{2} \right\rangle_2 + \left| \frac{1}{2}, -\frac{1}{2} \right\rangle_1 \otimes \left| \frac{1}{2}, \frac{1}{2} \right\rangle_2 \right)$$

for $|0,0\rangle$:
• Its antisym to $|1,0\rangle$
• Its orthogonal to $|1,0\rangle$

So,

$$|0,0\rangle = \frac{1}{\sqrt{2}} \left(\left| \frac{1}{2}, \frac{1}{2} \right\rangle_1 \otimes \left| \frac{1}{2}, -\frac{1}{2} \right\rangle_2 - \left| \frac{1}{2}, -\frac{1}{2} \right\rangle_1 \otimes \left| \frac{1}{2}, \frac{1}{2} \right\rangle_2 \right)$$

and verify

$$\hookrightarrow \langle 1,0 | 0,0 \rangle = 0$$

$$\Rightarrow \underline{S^2 |0,0\rangle = 0} :$$

$$\begin{aligned} S^2 &= (S^{(1)} + S^{(2)})^2 = (S^{(1)})^2 + (S^{(2)})^2 + 2 \cdot S^{(1)} \cdot S^{(2)} \\ &= (S^{(1)})^2 + (S^{(2)})^2 + S_+^{(1)} S_-^{(2)} + S_-^{(1)} S_+^{(2)} \\ &\quad + 2 S_z^{(1)} S_z^{(2)} \end{aligned}$$

$$\hat{J}^{(1)} \cdot \hat{J}^{(2)} = \frac{1}{2} \left(J_+^{(1)} J_-^{(2)} + J_-^{(1)} J_+^{(2)} \right) + J_z^{(1)} J_z^{(2)}$$

$$S^2 |0,0\rangle = \frac{3}{2} \hbar^2 |0,0\rangle + \left(S_+^{(1)} S_-^{(2)} + S_-^{(1)} S_+^{(2)} + 2 S_z^{(1)} S_z^{(2)} \right) |0,0\rangle$$

$$\rightarrow 2 S_z^{(1)} S_z^{(2)} = 2 \left(\frac{\hbar}{2} \right) \left(-\frac{\hbar}{2} \right) |0,0\rangle = -\frac{\hbar^2}{2} |0,0\rangle$$

$$\rightarrow S_{\pm} \left| \frac{1}{2}, \mp \frac{1}{2} \right\rangle = \hbar^2 \left| \frac{1}{2}, \pm \frac{1}{2} \right\rangle$$

Thus we get $S^2 |0,0\rangle = 0$

Summarizing :

$$1. |1,1\rangle = \left| \frac{1}{2}, \frac{1}{2} \right\rangle_1 \otimes \left| \frac{1}{2}, \frac{1}{2} \right\rangle_2$$

$$2. |1,0\rangle = \frac{1}{\sqrt{2}} \left(\left| \frac{1}{2}, \frac{1}{2} \right\rangle_1 \otimes \left| \frac{1}{2}, -\frac{1}{2} \right\rangle_2 + \left| \frac{1}{2}, -\frac{1}{2} \right\rangle_1 \otimes \left| \frac{1}{2}, \frac{1}{2} \right\rangle_2 \right)$$

$$3. |1,-1\rangle = \left| \frac{1}{2}, -\frac{1}{2} \right\rangle_1 \otimes \left| \frac{1}{2}, -\frac{1}{2} \right\rangle_2$$

$$4. |0,0\rangle = \frac{1}{\sqrt{2}} \left(\left| \frac{1}{2}, \frac{1}{2} \right\rangle_1 \otimes \left| \frac{1}{2}, -\frac{1}{2} \right\rangle_2 - \left| \frac{1}{2}, -\frac{1}{2} \right\rangle_1 \otimes \left| \frac{1}{2}, \frac{1}{2} \right\rangle_2 \right)$$

Hence we found out $\alpha, \beta, \gamma, \delta$.

Briefer Notation :

$$|1,1\rangle = |\uparrow\uparrow\rangle$$

$$|1,0\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$$

$$|1,-1\rangle = |\downarrow\downarrow\rangle$$

$$|0,0\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$$

Feynman - Hellman lemma :

Hamiltonian : $H(\lambda)$

Parameter : λ

Normalised energy eigenstate : $\psi(\lambda)$

Energy : $E(\lambda)$

$$\text{Then : } \frac{dE(\lambda)}{d\lambda} = \left\langle \psi(\lambda) \left| \frac{dH(\lambda)}{d\lambda} \right| \psi(\lambda) \right\rangle$$

or

$$\Delta E = \langle \psi(0) | \Delta H | \psi(0) \rangle : \text{first order in}$$

ΔH is computed in "unperturbed" state. ^{perturbation}

TODO: where/when to apply the lemma.

